

Can Machine Learning Really Predict the Stock Market?

A Plain-Language Critical Study of Retail Stock-Prediction Pipelines

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Case studies: technology equities and NFLX (2014–2023)

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Abstract—Plenty of online projects claim to predict stock prices, and many of them look almost perfect. They are not. The usual setup asks a model to guess tomorrow’s price, but tomorrow’s price is nearly the same as today’s, so the model just repeats today’s number and scores brilliantly without actually predicting anything. We tested this properly on ten years of real Netflix data. A one-line “tomorrow = today” rule beat a model trained on thirteen indicators, and once we asked the honest question, whether the next day goes up or down, accuracy fell below a coin flip. The same data gives $R^2 = 0.996$ under a sloppy test and roughly zero under a careful one, so the score was really measuring the test, not the model. Prediction does work in a couple of places, guessing how big a move will be and comparing stocks against each other, just not the way most beginners hope for. The reason runs deep: a market is not like the weather. When someone spots a pattern and trades on it, the trade moves the price and the pattern disappears.

Index Terms—financial machine learning, data leakage, cross-validation, non-stationarity, baselines, efficient markets

I. THE QUESTION WE STARTED WITH

USING past prices and a few familiar indicators, moving averages, RSI, and the like, to guess where a stock goes next: that is the pitch behind thousands of online notebooks. A lot of them show a prediction line that sits almost exactly on top of the real price, with tiny error, and offer it as proof that the market can be forecast.

It is not proof, and this paper explains why. We show what goes wrong, test the idea properly on real data, and then look at where prediction actually holds up. The language stays simple throughout, and the mistakes line up with the warnings in López de Prado’s book [1].

II. WHY THE POPULAR MODELS LOOK AMAZING (AND WHY IT IS FAKE)

The trap is almost always the same one. The model is asked to predict tomorrow’s price, which sounds reasonable until you notice how little prices actually move.

From one day to the next a stock usually shifts by around a percent, so tomorrow’s price is very close to today’s. A model that simply says “tomorrow’s price = today’s price” is already right nearly every time. It has forecast nothing, yet the usual

R^2 score looks superb, because the guess and the answer sit so close together.

The point is easier to see written out:

$$\text{tomorrow's price} = \underbrace{\text{today's price}}_{\text{already known}} + \underbrace{\text{the change}}_{\text{the hard, real question}}. \quad (1)$$

Predicting the price lets the model coast on the part you already know. The bit that actually needs forecasting is the change, and the change is mostly noise.

Figure 1 makes this concrete on data that even includes three crashes, so nobody can call it too tidy. The top panel is the price. It looks smooth and easy to follow, and “copy today” scores $R^2 = 0.9985$. The middle panel is the very same data written as daily changes, and it dissolves into static. The bottom-right panel settles the matter: plot today’s move against tomorrow’s and you get a formless blob. There is nothing there to learn.

III. THE HONEST TEST ON REAL DATA (NETFLIX, 2014–2023)

We checked all this on fresh numbers: ten years of daily Netflix (NFLX) prices with thirteen ready-made indicators. This time we did the three things the tutorials leave out. We compared against a trivial baseline, we predicted the return instead of the price, and we tested without letting the model see the future. Table I has the numbers, and they do not leave much room for interpretation.

The trivial baseline wins. “Tomorrow = today” lands at an RMSE of \$11.94. A linear model with all thirteen indicators actually does worse, at \$16.21. It never learned to forecast; it learned a slightly clumsier version of “copy yesterday.” Figure 3 shows the three lines lying on top of one another.

Ask the real question and the skill vanishes. Switch the target to the next-day return and the out-of-sample R^2 turns negative (−0.023), with direction called correctly less than half the time (49.4%), a shade worse than flipping a coin. A heavier random-forest model does no better, at 48.2%.

The test method can invent the result on its own. Push the same data through a shuffled split and it proudly reports

Why 'predicting the price' is easy and useless, but 'predicting the change' is hard

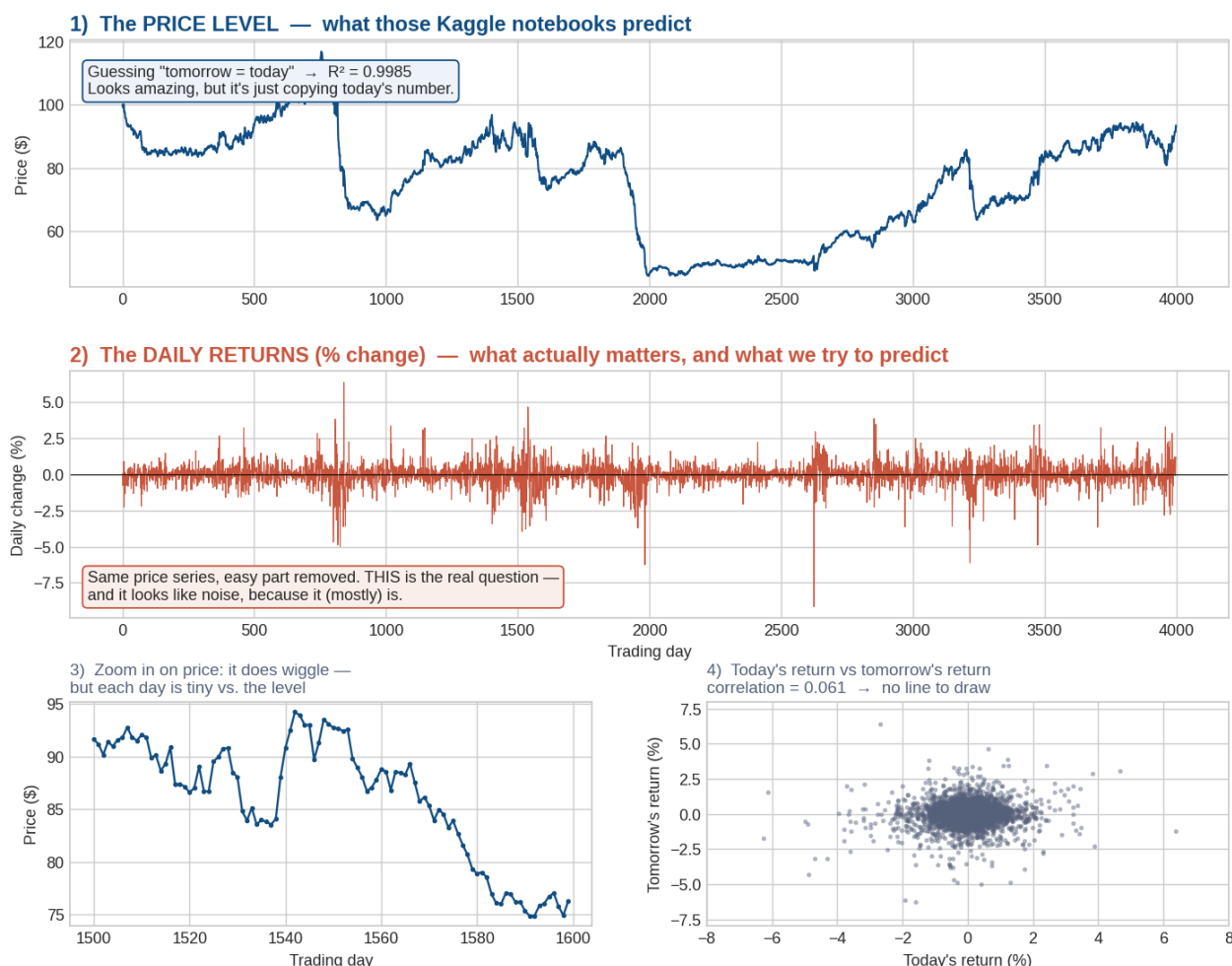


Fig. 1. The whole problem in one picture. The price (top) looks smooth and predictable, so “tomorrow = today” scores almost a perfect R^2 . The same data as daily changes (middle) is just noise, and today’s change tells you nothing about tomorrow’s (bottom right: a formless blob). Predicting the price is easy and useless; predicting the change is the real, hard task.

$R^2 = 0.996$. Run a careful, time-respecting split and it drops to around zero or below. That headline number was coming from the way the test was set up, not from anything the model knew (Figure 2).

IV. THE THREE MISTAKES THAT FOOL ALMOST EVERYONE

Both notebooks we looked at, an Apple regression and a Netflix LSTM, trip over the same three things.

First, predicting the price rather than the change. That is what lets the model coast on today’s number. It is better to predict the return, or just whether the stock closes up or down.

Second, letting the future bleed into the past. It happens two ways: scaling the data on the whole set before splitting off a test period, or shuffling time-ordered rows so tomorrow ends up teaching the model about today. Either one flatters the

TABLE I
HONEST NETFLIX EXPERIMENT. THE TRIVIAL BASELINE IS NOT BEATEN; THE REAL TARGET GIVES NEAR-ZERO OR NEGATIVE SKILL; AND A LEAKY TEST INVENTS A HIGH SCORE.

What we tried	Measure	Result
“Tomorrow = today” (trivial baseline)	RMSE	\$11.94
Linear model, all indicators (price)	RMSE	\$16.21
Linear model, all indicators (price)	R^2	0.971
Shuffled (leaky) test, price	R^2	0.996
Linear model, next-day return	R^2	−0.023
Linear model, next-day return	direction	49.4%
Random forest, next-day return	direction	48.2%
Honest time-respecting test, return	R^2	−0.092

score. The fix is to split by time and to prepare the data using the training rows only.

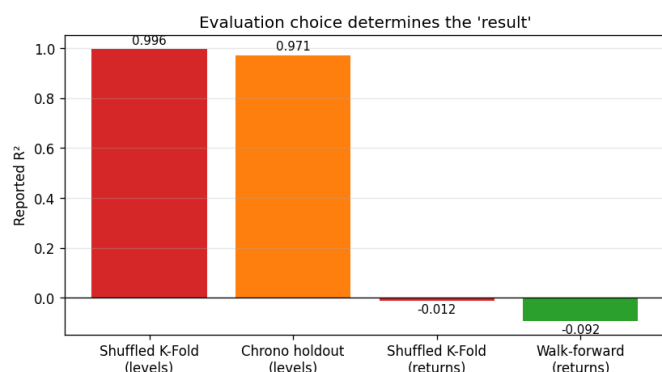


Fig. 2. The same Netflix data, four different test methods. The two leaky/optimistic methods (left) report near-perfect scores. The two honest methods (right) reveal the truth: no real skill. How you test decides the “result.”



Fig. 3. Predicting the Netflix price level. The model line, the “tomorrow = today” line, and the real price are impossible to tell apart. The “good” result is just the model echoing yesterday’s price.

Third, skipping the baseline. A score on its own says nothing. “My error is only \$5” sounds impressive right up until you find out “copy today” matches it. Always hold the model against “tomorrow = today,” “always guess up,” and plain buy-and-hold.

None of this is our own invention; these are the standard cautions in López de Prado [1]. He goes further, with stationary features, smarter labels like the triple-barrier method, and leak-free cross-validation, but the three above are enough to dissolve the illusion.

V. WHERE PREDICTION ACTUALLY WORKS

So markets are not a lost cause everywhere. It is that the next-day direction of a single stock is about the barest patch of ground you could pick. There are two better spots.

A. The size of the move (volatility)

You cannot say which way a stock will go tomorrow, but you can say a fair amount about how far it will travel. Wild days bunch together, and so do quiet ones. In the real data we looked at, the direction of returns carried almost no memory from one day to the next (correlation -0.003), while the size of the moves clearly did (0.25). That is why volatility, not direction, is the target where an honest model can actually outrun a dumb one, and it is what risk and options desks care about anyway.

B. Comparing stocks against each other

Most of what a single stock does on a given day is just the market as a whole rising or falling, one big wave that is very hard to call. Ask a relative question instead, like whether Apple beats Microsoft, and that wave hits both names and mostly cancels out. You are left with a cleaner read on the two companies.

We tried this on five years of real S&P 500 data. On a typical day two stocks move together at a correlation of 0.28, the wave. Apple on its own follows the market at 0.44. But “Apple minus Microsoft” follows it at just -0.08 ; the wave has washed out. Build that into a “buy the winners, short the losers” portfolio and the whole thing barely tracks the market at all (correlation -0.11 ; Figure 4, right). It is market-neutral: it does not much mind whether the market goes up or down.

An honest caveat. Over this stretch, a strong bull market, the market-neutral version actually earned less than simply holding the market (5.5% a year against 12.2%). Its point is not a bigger return, it is a return that is disconnected from the market, which is handy for safety and would likely earn its keep in a downturn rather than a rally. On top of that, this quick test only includes firms that stayed in the S&P 500, which quietly flatters the result. So read it as a demonstration of the idea, not a recipe for making money.

VI. WHY THE SIGNAL IS SO WEAK: THE REAL REASON

There is a fair objection here. Traders use these indicators and make money, and an AI can chew through far more data than any person, so why can it not just win? Three honest answers.

A. “Traders use indicators and profit” does not hold up

Most small traders do not beat buy-and-hold; the large studies of real brokerage accounts find the majority fall behind it [4]. The winners are the ones you hear about, which is a biased sample, because nobody posts about the account they quietly closed. And the traders who do stick around usually make their money elsewhere: tight risk control, sensible position sizing, market-making that earns the spread between buy and sell, or getting paid to carry risk. The edge is in how they bet, not in what they predict.

B. More compute does not conjure a signal

AI cleans up at chess and protein folding because those have fixed rules and a strong, steady pattern waiting to be found. Markets are not like that. For one thing, the pattern fights back: find something that works, trade it, and the trading itself nudges the price until the edge is gone. A signal that evaporates the moment you use it is not something you can brute-force. For another, there is barely any signal to begin with, as we measured directly. Point a bigger model at noise and it will not uncover hidden structure; it will invent patterns that fall apart later. Here extra capacity hurts rather than helps.

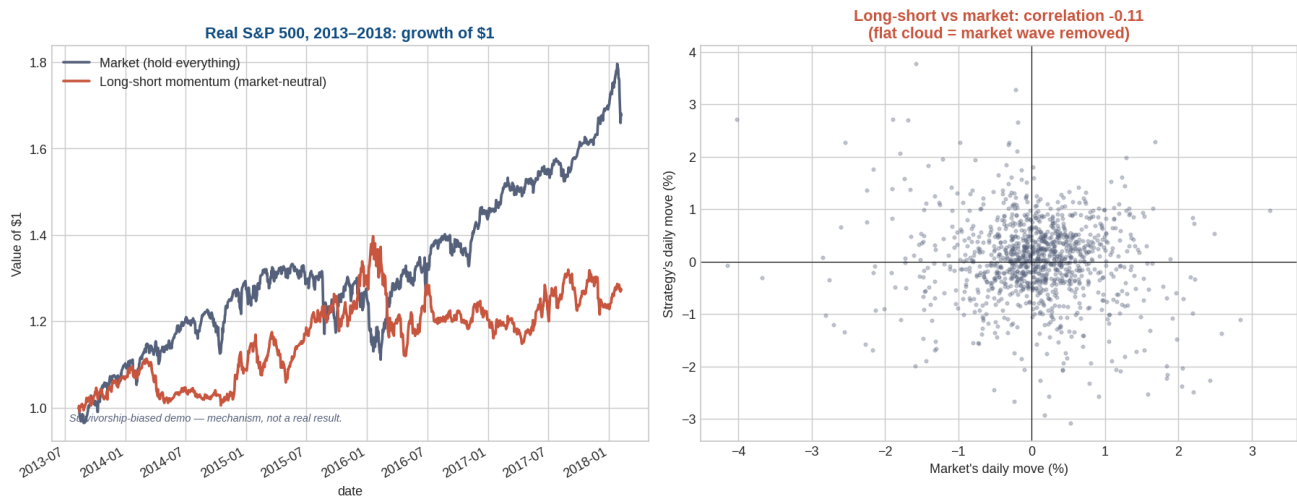


Fig. 4. Comparing stocks cancels the market wave (real S&P 500, 2013–2018). Left: growth of \$1 for holding the market vs. a market-neutral “winners minus losers” strategy. Right: the strategy’s daily moves against the market’s form a flat cloud (correlation -0.11), so the unpredictable market wave has been removed.

C. Reading the news is real, but the news is already in the price

Trading on headlines, filings, even satellite photos, is a genuine business. The catch is that the obvious reading is not an edge. When a shipping lane shuts, thousands of algorithms reach the same “oil up, airlines down” conclusion within seconds, and the price has moved before a slower player can lift a finger. What is left to win on is speed (an expensive arms race), a subtler angle the crowd missed, or information nobody else has. Every one of those is an advantage in information or infrastructure, not sharper thinking about the same public indicators. Grossman and Stiglitz made the point long ago [5]: if prices already reflected everything, no one would bother paying to dig up information, so the surviving edge is always small and short-lived.

D. A market is a competition, not a forecast

This is what ties it together. The weather does not change because you predicted it; a price does, because people act on the prediction. Forecasting a market changes the very thing you are forecasting. “Just add more AI and data” buys you nothing on its own, since everyone else added it too and the advantage gets bid away until it is worth about what the technology costs to run. For the single-stock, public-indicator, next-day model here, whatever edge remains is, by our measurements, indistinguishable from zero.

VII. WHAT TO DO INSTEAD: A SHORT CHECKLIST

- **Predict a change, not a price.** Returns or up/down, never the level.
- **Always state a baseline.** If you cannot beat “tomorrow = today,” “always up,” and buy-and-hold, you have not shown anything.
- **Split by time, never shuffle.** And prepare the data on the training rows only.

- **Distrust a beautiful result.** A perfect-looking chart usually means a leak or a bug, not brilliance.
- **Aim where signal lives.** Volatility and stock-versus-stock bets beat next-day direction.

VIII. CONCLUSION

Most of the impressive stock-price predictions floating around online are not forecasts at all. They are today’s price wearing a disguise, propped up by a leaky test. Test the idea honestly on real data and the next-day move of a single stock turns out to be essentially unpredictable from the usual indicators, with a one-line baseline beating the fancy model. That is not a failure. It is the right answer, and knowing why, because markets compete the easy patterns away, is worth a great deal more than a fake perfect score.

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